

Probing the Dark Universe

Machine Learning and Statistical Approaches to Gamma-Ray
Dark Matter Searches

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Thesis submitted for the degree of
Doctor of Philosophy

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February 6, 2026

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ABSTRACT

This thesis explores...

INTRODUCTION

The abstract

THE DARK MATTER PROBLEM

The beginning [1]

THE GAMMA-RAY SKY AND FERMI-LAT

This chapter describes...

**STATISTICAL METHODS AND MACHINE LEARNING IN
ASTROPHYSICS**

This chapter describes...

THE GALACTIC CENTER EXCESS (GCE)

This chapter describes...

SEARCHING FOR DARK MATTER SUBSTRUCTURES

This chapter describes...

FROM INDIVIDUAL SOURCES TO POPULATIONS

This chapter describes...

PROBABILISTIC CATALOGING

This chapter describes...

CROSS-CORRELATIONS AND FUTURE PROSPECTS

This chapter describes...

SUMMARY AND OUTLOOK

In this thesis...

BIBLIOGRAPHY

- [1] M. Fornasa et al., *Angular power spectrum of the diffuse gamma-ray emission as measured by the Fermi Large Area Telescope and constraints on its dark matter interpretation*, *Phys. Rev. D* **94** (2016) 123005 [1608.07289].